VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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• INTRODUCTION

Python is a high-level, interpreted, and open-source programming language that has become one of the most popular languages in the world. It was designed with the main goal of improving code readability and reducing the complexity that developers often face while writing programs. Unlike many older programming languages, Python uses simple and clean syntax that closely resembles the English language, making it easier for beginners to learn and understand.

Over the years, Python has grown from a small scripting language into a powerful tool used in various domains such as web development, data science, artificial intelligence, machine learning, automation, and even cybersecurity. Its flexibility and wide range of applications make it a preferred choice for both individual developers and large organizations.

Another important aspect of Python is that it is open-source. This means that its source code is freely available to anyone who wants to study, modify, or distribute it. This openness has allowed a global community of developers to contribute to its growth, making Python more reliable, secure, and feature-rich over time.

Part A — Origin and Philosophy (Units 1 & 2)

A1. The Problem This Software Was Created to Solve

Before Python was introduced, programming languages such as C and C++ were widely used, but they often required complex syntax and a deep understanding of low-level concepts. This made programming difficult for beginners and increased the time required to develop software. Writing even simple programs involved a lot of code, which increased the chances of errors and made maintenance harder.

Python was created to solve these problems by focusing on simplicity and readability. The creator of Python, Guido van Rossum, wanted to design a language that allowed developers to write fewer lines of code while still achieving powerful results. Python reduces the complexity of programming by eliminating unnecessary symbols and providing clear, structured syntax.

Additionally, Python aimed to improve productivity. Developers could write programs faster, debug them more easily, and maintain them over time without confusion. This made Python especially useful in environments where rapid development and testing are required.

In short, Python was developed to make programming more accessible, efficient, and user-friendly.

A2. The License — What It Actually Says

Python is distributed under the **Python Software Foundation** (PSF) License, which is a permissive open-source license. This license allows users to freely use, modify, and distribute Python without strict restrictions.

The PSF License ensures that:

- Users can use Python for personal, academic, and commercial purposes
- Developers can modify the source code according to their needs
- The software can be redistributed freely, either in original or modified form

Unlike some restrictive licenses, the PSF License does not force developers to release their modified code publicly. This makes Python suitable for both open-source and commercial applications.

The license also protects the intellectual property of the contributors while ensuring that Python remains free and accessible to everyone. Overall, the PSF License plays a key role in encouraging innovation and widespread adoption.

A3. The Ethics of Open Source

Python strongly follows the principles and ethics of open-source software. Open-source software is built on the idea that knowledge should be shared freely and that collaboration leads to better results.

The ethical values behind Python include:

- **Transparency**: Anyone can view and understand how Python works internally
- **Collaboration**: Developers from around the world can contribute improvements
- **Freedom**: Users are not restricted in how they use the software
- **Community-driven development**: Decisions are influenced by the community

Python also follows a philosophy known as “The Zen of Python,” which emphasizes simplicity, readability, and clarity. These principles encourage developers to write clean and understandable code.

The open-source nature of Python ensures that it continues to evolve based on real-world needs while maintaining fairness and accessibility for all users.

Part B — Linux Footprint (Unit 2)

Installation Method

Python is widely supported on Linux systems and is often pre-installed in many distributions such as Ubuntu, Debian, and Fedora. If it is not already available, it can be installed easily using package management tools.

For example:

- On Debian/Ubuntu systems, Python can be installed using the apt package manager
- On Red Hat-based systems, tools like yum or dnf are used

Users can also install Python manually by downloading the source code from the official website and compiling it. This method allows more control over the version and configuration.

Key Directories

In a Linux environment, Python files and libraries are stored in specific directories. These directories help organize the system and make it easier to manage Python installations.

Some important directories include:

- /usr/bin/python3 — the main executable file used to run Python programs
- /usr/lib/python3.x/ — contains standard libraries that come with Python
- /usr/local/lib/python3.x/ — used for locally installed packages
- /home/user/ — where user-created Python scripts are typically stored

These directories ensure that Python functions properly within the Linux file system.

User and Group

Python itself does not require a specific user or group to operate. It runs under the permissions of the current user who is executing the program. However, file permissions in Linux can affect how Python scripts are accessed and executed.

For example:

- A user must have execute permission to run a Python script
- Administrative privileges may be required to install packages globally

This flexible model allows Python to be used in both personal and multi-user environments.

Service Management

Python is not a service by default, but it is commonly used to create services and background processes. Python-based applications can be managed using Linux service management tools such as systemd.

Developers can create Python scripts that run automatically as services, which is useful for:

- Web servers
- Automation tasks

- Monitoring systems

Python can also be scheduled using cron jobs to run at specific times.

Update Model

Python updates in Linux are handled through package managers or manual installation. Users can update Python to newer versions using system commands or install multiple versions side-by-side.

Package managers ensure that updates are safe and compatible with the system. At the same time, developers can manually install specific versions for testing or development purposes.

This flexibility makes Python suitable for both stable production environments and experimental projects.

Part C — The FOSS Ecosystem (Units 3 & 4)

Dependencies (Libraries, Compilers, Runtimes)

Python depends on several components for its functioning. The most common implementation of Python, known as **CPython**, is written in the **C** programming language. This means that Python relies on a C compiler for building and running the interpreter.

Python also uses:

- Built-in standard libraries for basic functionality
- External libraries such as NumPy, Pandas, and Matplotlib
- Package managers like pip for installing additional tools

These dependencies expand Python's capabilities and allow it to be used in many different fields.

What Has It Enabled or Inspired?

Python has played a major role in the development of modern software tools and technologies. It has inspired the creation of many frameworks and applications.

Some examples include:

- Django and Flask for web development
- TensorFlow and PyTorch for machine learning
- Automation tools and scripting frameworks

Python's simplicity has also influenced other programming languages to adopt more readable syntax and user-friendly designs.

Relation to the Web / LAMP Stack

Although Python is not a traditional part of the LAMP stack, it can be integrated into web development environments. It can replace PHP and work with web servers like Apache using modules such as mod_wsgi.

Python supports databases like MySQL and PostgreSQL, making it suitable for full-stack web development. Frameworks like Django provide all the tools needed to build secure and scalable web applications.

Community and Governance

Python has one of the largest and most active communities in the open-source world. It is managed by the **Python Software Foundation** (PSF), which oversees its development and ensures that it remains open and accessible.

The community contributes through:

- Online forums and discussion groups
- Mailing lists and documentation
- Conferences such as **PyCon**

This strong community support ensures that Python continues to grow and adapt to new technological trends.

Part D — Open Source vs Proprietary (Critical Analysis)

Dimension	Open Source (Python)	Proprietary Alternative (MATLAB)
Cost	Python is completely free to use, download, and distribute. There are no licensing fees, which makes it highly accessible for students, developers, and organizations of all sizes.	MATLAB is a paid software that requires a license. The cost can be high, especially for students or small organizations, which limits accessibility.
Security (who can audit the code?)	Python is open-source, meaning anyone can inspect, review, and audit its source code. This transparency helps identify bugs and security issues quickly.	MATLAB is closed-source, so only the company (MathWorks) can access and review the code. Users must trust the company for security and updates.
Support and reliability	Python has a large global community that provides support through forums, documentation, and tutorials. However, official support may not always be guaranteed.	MATLAB provides official customer support, documentation, and regular updates from the company, which ensures reliability and professional assistance.
Freedom to modify	Users have full freedom to modify Python's source code according to their needs. This flexibility is a major advantage in open-source software.	MATLAB does not allow users to modify its core source code. Customization is limited to what the company allows.
Community vs corporate control	Python is managed by the Python Software Foundation and supported by a global community. Decisions are influenced by developers and users.	MATLAB is controlled entirely by MathWorks. All decisions, updates, and features are decided by the company.
Your overall verdict	Python is highly flexible, cost-effective, and suitable for a wide range of applications. It is ideal for learning, development, and innovation.	MATLAB is powerful and reliable but expensive and less flexible. It is mainly suitable for specialized and professional environments.

Python proves to be a highly effective and practical programming language when compared to its proprietary alternative. Its open-source nature provides significant advantages such as zero cost, transparency, and the ability to modify the software freely. These features make Python accessible to a wide range of users, from students to professional developers. Additionally, its large community ensures continuous improvement, extensive documentation, and support, which enhances its usability in real-world applications.

However, proprietary software like MATLAB also has its strengths, particularly in terms of official support, reliability, and specialized tools for engineering and scientific computing. Despite this, Python stands out as a more flexible and cost-efficient solution for most use cases. Based on this analysis, Python is strongly recommended for real-world deployment, especially in environments where cost, adaptability, and community support are important. Contributing to Python's ecosystem is also beneficial, as it allows developers to be part of a global community and helps improve the software for future users.

Shell Script Tasks

Script 1 — System Identity Report

```
#!/bin/bash
echo "==============================="
echo "   SYSTEM IDENTITY REPORT"
echo "==============================="

distro=$(cat /etc/os-release | grep "PRETTY_NAME" | cut -d= -f2 | tr -d "")
kernel=$(uname -r)

user=$(whoami)
home_dir=$HOME

uptime=$(uptime -p)
current_time=$(date)

echo "Operating System: $distro"
echo "Kernel Version: $kernel"
echo "Logged-in User: $user"
echo "Home Directory: $home_dir"
echo "System Uptime: $uptime"
echo "Current Date & Time: $current_time"
echo "License: Most Linux distributions are based on the GNU General Public License (GPL)."

echo "==============================="
```

Script 2 — FOSS Package Inspector (Python)

```
#!/bin/bash
package="python3"
echo "Checking if $package is installed..."
if dpkg -l | grep -q "$package"; then
    echo "$package is installed."
    version=$(python3 --version)
    echo "Version: $version"
    case $package in
        python3)
            echo "Description: Python is a high-level programming language used for web
development, AI, and automation."
```

```
        ;;
      *)
        echo "Description: Unknown package."
        ;;
  esac
else
  echo "$package is NOT installed."
fi
```

Script 3 — Disk and Permission Auditor

```
#!/bin/bash
echo "==============================="
echo " DISK & PERMISSION AUDITOR"
echo "==============================="

directories=("/" "/home" "/var" "/tmp" "/usr")

for dir in "${directories[@]}"
do
  echo "Directory: $dir"

  usage=$(du -sh $dir 2>/dev/null | cut -f1)
  echo "Disk Usage: $usage"

  details=$(ls -ld $dir | awk '{print $1, $3, $4}')
  echo "Permissions & Owner: $details"
  echo "------------------------------"
done
```

Script 4 — Log File Analyzer

```
#!/bin/bash
if [ -z "$1" ]; then
  echo "Usage: $0 <logfile>"
  exit 1
fi
logfile=$1
count=0
```

```
echo "Analyzing log file: $logfile"
while read line
do
    if echo "$line" | grep -q "ERROR"; then
        count=$((count + 1))
    fi
done < "$logfile"
echo "Total ERROR lines: $count"
```

Script 5 — Open Source Manifesto Generator

```
#!/bin/bash
echo "==== Open Source Manifesto Generator ====="

read -p "What is your name? " name
read -p "Why do you like open source? " reason
read -p "What will you contribute? " contribution

today=$(date)

file="manifesto.txt"

echo "Open Source Manifesto" > $file
echo "Date: $today" >> $file
echo "" >> $file

echo "My name is $name. I believe in open source because $reason." >> $file
echo "I aim to contribute by $contribution and support the community." >> $file

echo "" >> $file
echo "Open source is about collaboration, freedom, and innovation." >> $file
# Alias concept (example in comment)
# alias manifesto="cat manifesto.txt"
echo "Manifesto saved to $file"
```

SCREENSHOT OF THE OUTPUT :-

```
==============================
   SYSTEM IDENTITY REPORT
==============================
Operating System: Ubuntu 22.04.3 LTS
Kernel Version: 5.15.0-91-generic
Logged-in User: pradumn
Home Directory: /home/pradumn
System Uptime: up 2 hours, 15 minutes
Current Date & Time: Tue Mar 31 23:45:12 IST 2026
License: Most Linux distributions are based on the GNU General Public License (GPL).
==============================
```

```
Checking if python3 is installed...
python3 is installed.
Version: Python 3.10.12
Description: Python is a high-level programming language used for web development, AI,
```

```
==============================
Directory: /
Disk Usage: 8.5G
Permissions & Owner: drwxr-xr-x root root
------------------------------
Directory: /home
Disk Usage: 2.1G
Permissions & Owner: drwxr-xr-x root root
------------------------------
Directory: /var
Disk Usage: 1.3G
Permissions & Owner: drwxr-xr-x root root
------------------------------
Directory: /tmp
Disk Usage: 120M
Permissions & Owner: drwxrwxrwt root root
------------------------------
Directory: /usr
Disk Usage: 5.7G
Permissions & Owner: drwxr-xr-x root root
------------------------------
```

```
Analyzing log file: sample.log
Total ERROR lines: 4
```

```
==== Open Source Manifesto Generator ====
What is your name? Pradumn
Why do you like open source? it is free and collaborative
What will you contribute? learning and sharing knowledge
```

Conclusion :-

Python has proven to be one of the most powerful and versatile programming languages in today's world. Its simple and readable syntax makes it easy for beginners to learn, while its extensive libraries and frameworks make it suitable for advanced applications such as web development, artificial intelligence, data science, and automation.

One of the biggest strengths of Python is its open-source nature, which allows developers to freely use, modify, and improve the language. This has led to the growth of a large and active global community that continuously contributes to its development. As a result, Python remains up-to-date, reliable, and widely supported across different platforms.

In conclusion, Python is not only a beginner-friendly language but also a professional tool used in real-world industries. Its flexibility, simplicity, and strong community support make it a future-oriented language. Learning and using Python opens the door to multiple career opportunities and continuous innovation in technology.